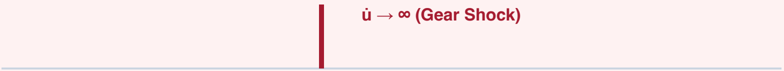
COMPARISON OF HARD STEP TAKEOVER VS. C^2 BUMPLESS TORQUE BLENDING

Eliminating motor current inrush spikes (7.4 A → 1.8 A) and infinite torque rate-of-change ($\dot{u}$) during human intervention

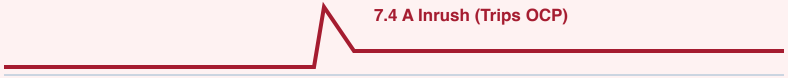
❌ HARD STEP SWITCHING (DISCONTINUOUS TAKEOVER)

(a) Commanded Torque u(t) — Discontinuous Step

(b) Torque Rate u̇(t) — Dirac Impulse δ(t)



(c) Phase Current I(t) — 7.4 A Inrush Spike

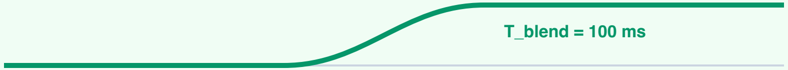


PHYSICAL HARDWARE DAMAGE:

- Inrush current trips inverter overcurrent fault (OCP).
- Instantaneous torque step shears harmonic drive teeth.
- Robot jerks violently against human operator's hand.

✓ C2 BUMPLESS BLENDING (SIGMOID INTERPOLATION)

(a) Commanded Torque u(t) — Smooth Sigmoid S-Curve



(b) Torque Rate u̇(t) — Bounded Bell Curve (≤ 11 Nm/s)

(c) Phase Current I(t) — Continuous Delivery (1.8 A)

PHYSICAL SYSTEMS ADVANTAGES:

- Zero inverter overcurrent trip risk; 100% uptime.
- Gearbox torque derivative strictly bounded within mechanical specs.
- Intuitive, smooth haptic feel for the human co-pilot.